ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET first paper. Section A (30): multiple choice, true or false, matching Python terms and labelling the Scratch window · Section B (70): computer hardware and software, Git, errors, binary numbers, a Scratch program to study, a vending-machine flowchart and a Python program with ten errors to fix.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Start each section on a new page.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which ONE of the following is system software?

A Microsoft Word B Excel C Windows 11 D Paint

1.2 Which ONE of the following is an input device?

A Scanner B Speaker C Printer D Projector

1.3 RAM is described as …

A permanent storage. B volatile memory.
C an output device. D an operating system.

1.4 CPU stands for …

A Computer Power Unit. B Central Program Utility.
C Control Processing Unit. D Central Processing Unit.

1.5 In the IPO model, typing marks into a program is part of the …

A input. B processing. C output. D storage.

1.6 Which flowchart symbol is used for input or output?

A Rectangle B Diamond C Parallelogram D Oval

1.7 Which Python function reads text typed by the user?

A print() B input() C len() D int()

1.8 Which Git command shows which files have been changed?

A git push B git clone
C git init D git status

1.9 In Scratch, the "repeat" and "if … then" blocks are found in the … category.

A Control B Looks C Sound D Motion

1.10 The binary number 1111 is equal to the decimal number …

A 14. B 15. C 16. D 17.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 Python is case sensitive, so Total and total are different names.

2.2 In Scratch, the "when green flag clicked" block starts a script.

2.3 A Python list cannot be changed after it has been created.

2.4 The # symbol starts a comment in Python.

2.5 An interpreter translates the whole program into machine code before running it.

(5)

[5]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

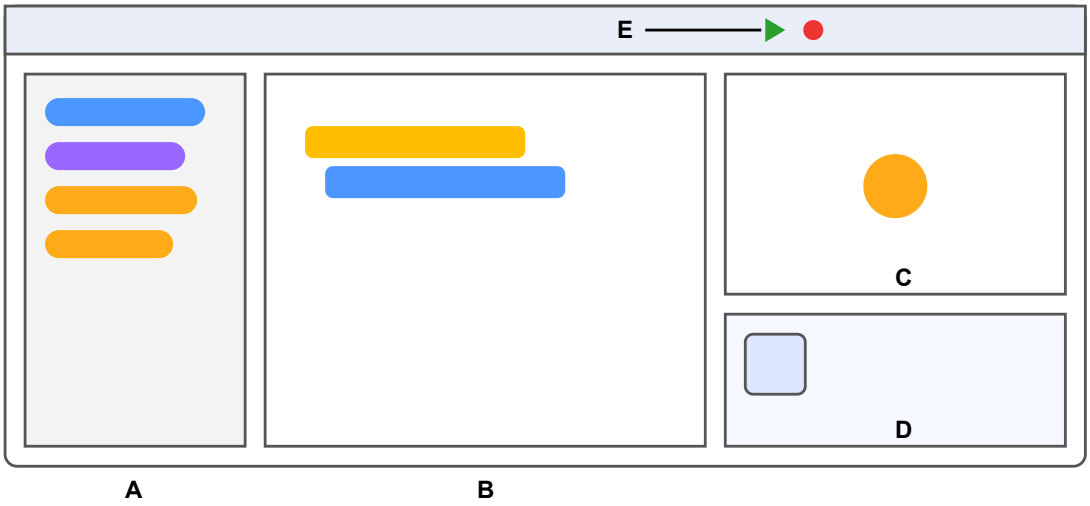
	COLUMN A		COLUMN B
3.1	Changes text such as "25" into a whole number	A	break
3.2	Adds an item to the end of a list	B	append()
3.3	Repeats code while a condition is true	C	range()
3.4	Gives another path when the if condition is false	D	int()
3.5	Data type that stores text	E	Boolean
3.6	Data type whose value is either True or False	F	type()
3.7	Produces a sequence of numbers for a for loop	G	while loop
3.8	Rounds a number to a given number of decimal places	H	string
3.9	Gives the data type of a value	I	round()
3.10	Leaves a loop immediately	J	else

(10)

[10]

QUESTION 4

The sketch below shows the main parts of the Scratch window. Name the parts labelled A–E. Write only the answer next to the letter in the ANSWER BOOK.



4.1 A

4.2 B

4.3 C

4.4 D

4.5 E (the green symbol)

(5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 Answer the following questions:

5.1.1 Which component stores files and programs permanently? (1)

5.1.2 What does a network interface card (NIC) do? (1)

5.1.3 Which type of port is used to connect a mouse, keyboard or flash drive? (1)

5.1.4 Which part keeps the CPU from overheating? (1)

5.2 Explain what a repository and a commit are in Git. (4)

5.3 Explain the difference between system software and application software, and give ONE example of each. (4)

5.4 Explain the difference between a syntax error and a logic error, and give ONE example of each. (4)

5.5 Give TWO advantages of using version control when a team works on the same program. (4)

5.6 Convert the decimal number 37 to binary. Show your working. (5)

[25]

QUESTION 6

Study the Scratch program below and answer the questions that follow.

```
when green flag clicked
set [battery] to 20
set [charges] to 0
repeat until < (battery) > 99 >
    change [battery] by 10
    change [charges] by 1
    say (join [Battery: ] (battery)) for 1 seconds
end repeat
if < (charges) > 5 > then
    say [Charging took a long time] for 2 seconds
else
    say [Fully charged quickly] for 2 seconds
```

6.1 Name the TWO variables, give their starting values and say what each one represents. (4)

6.2 How many times does the loop run? Show how you got your answer. (4)

6.3 Write down the last message said inside the loop and the final message said after the loop. (4)

6.4 Name the TWO control structures used after the variables are set, and explain why control structures are important in programming. (4)

6.5 If the battery started at 25, it would go above 100. Suggest a change to the program that makes sure the battery never goes above 100. (4)

[20]

QUESTION 7

You are asked to design the logic of a college vending machine. A student inserts money and selects a snack. If the money is enough, the machine gives the snack and any change. If the money is not enough, the machine shows "Insufficient funds" and returns the money.

Draw a flowchart that represents the following process:

7.1 The process starts when a student walks up to the machine. (2)

7.2 The student inserts money and selects a snack. (2)

7.3 The machine checks whether the money is at least the price of the snack. (4)

7.4 If the money is not enough, it displays "Insufficient funds", returns the money and the process ends. (3)

7.5 If the money is enough, the machine gives the snack and calculates and gives the change. (3)

7.6 The process ends. (1)

[15]

QUESTION 8

The Python program below should ask how many items a customer is buying, read the price of each item, add up the prices and show the total before and after 15% VAT. The program contains TEN errors (syntax and logic). Write down the line number and describe each error (for example: Line 3 – variable not declared) in the ANSWER BOOK.

```
 1  def add_vat(price)
 2      return price * 0.15
 3
 4  items = int(input("How many items? ")
 5  total = 0
 6  for i in range(items + 1):
 7      price = input("Price of item " + str(i + 1) + ": ")
 8      if price < 0:
 9      print("Price cannot be negative")
10      else
11          total = totl + price
12  print("Total before VAT: R" + total)
13  Print("Total with VAT: R", add_vat(total))
```

(10)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER